Preliminary Categorial Analysis of Bunan

Based on the PhD thesis

by Manuel Widmer, 2014

At University of Bern

— by Cem Bozşahin, May 2026, Ling 58B of Boğaziçi Linguistics

Concentrating on:

(I) verbal categories

(II) ergativity and Bunan

(III) The middle voice and the transitive suffix

(IV) Two relative clause strategies

- Bunan is Tibeto-Burman, a minority language in India.
- Spoken by 3500-4000 speakers in Lahaul, the northernmost region of the state Himachal Pradesh.
- Traditional homeland is the Gahr Valley.

I) Verbal inflection (p. 478)

* 1) Tense – in almost all finite forms of the verb.

2) Mood – in small number of contractions

* 3) Conjunct–disjunct – robust

4) Evidentiality – not in the present tense

5) Person – virtually absent in young speakers

* 6) Number – robust

1) Forms of the verb (p. 478-490)

		INTR	MID	TR
– Finite form		V-ek	V-ek	V-kata (FUT)
– The infinitive		V-men	V-ɕom	V-tɕ-om
– The supine		V-de	V-ɕ-a	V-tɕ-a
– Participle	progressive	V-ka	V-ɕ-ka	V-ka (SG)
	active	V-i	V-ɕ-i	V-tɕ-i
– The converb		V-tɕʰi	V-ɕ-tɕi	V-∅-dʑi (PL)

aka. INF of purpose

3) The conjunct-disjunct system on the verb (p. 501 f)

- Conjunct (SG) ending V-ek : the speaker acts as the willful instigator of an event

- Disjunct (SG) ending V-are : the speaker does not assume this role.

(604) gi len liktpek.
1SG work do-TR-PRS.CJ.SG 'I am working.'

(605) gi duktpi gjolgare.
1SG pain=DAT=ABL recover-INTR-PRS.DJ.SG 'I am recovering from the sickness.'

6) <u>Number :</u> indicates plurality of the subject on the verb.

- number agreement is robust. (losing ground in recent past)
- Only animate subjects agree (p. 574)
- (CB: Same as Arabic, where both person and number agreements are for animate participants).

II) Ergativity and Burman

- Not abs-erg language or nom-acc language (p.743)
- Robust evidence for accusative alignment, and case marking reminiscent of split-ergativity (p.744)
- There seems to be no role asymmetry in its constructions.
- So what are the case markers?

ABS, ERG, DAT

Case marking and grammatical roles (Table 121, p.714)

	S	A	Ex	(O) J	St	T	R (Goal of a tv)
ABS	✓	✓		✓	✓	✓	
ERG		✓	✓				
DAT			✓	✓			✓

: accusative evidence

: ergative evidence, split

↑ J of a tv

Evidence of GRs from relative clauses? (p. 732)

- RCs derived from the active participle -i plus
  - pa the agentive nominalizer
  - tsuk the relativizing nominalizer (also infinitive rel. p. 755)
- Pa may relativize S, A and Ex arguments. (p. 733)
- tsuk may relativize S, A and P (p. 734)
- Seems neither syntactically ergative nor accusative.

Complement clauses (p. 788+) as evidence for GRs

- 5 types
- Supine form on the complement clause is most common (i.e., the purposive infinitive) (p. 788)
- p. 790–794 examples seem to suggest that the -SUP marked clause may target (A, S, Ex)
- Seems like accusativity – not strong evidence

going back to Widmer's assesment (p. 744)

- robust evidence for acc tendency in constuctions
- case marking split ergative

Basic order of constituents (p. 66?) in a finite matrix clause

1) Theme Compl Copula $\quad$ $S\backslash PredP\backslash NP : \lambda x\lambda p.\ be(px)$

2) UA/Ex verb $\quad$ $S\backslash NP : \lambda x.v'x$ $\quad$ UA/Ex

3) UA/Ex ʒ/St verb $\quad$ $S\backslash NP\backslash NP : \lambda x\lambda y.v'xy$ $\quad$ ʒ UA

4) UA R/T Theme verb $\quad$ $S\backslash NP\backslash NP\backslash NP : \lambda x\lambda y\lambda z.\ v'yxz$ $\quad$ UA R/T

∴ NO SIGN OF STRUCTURAL ERGATIVITY OR ACCUSATIVITY

II) The middle voice (p. 454)

Kemmer (1993): what the reflexive does to transitives, the middle voice does to intransitives (ie, they are semantically transitive)

English:

John exhausted himself.

ref

John grew weary.

m

Bunan marks the distinction on the verb forms.

- English example is not very clear.
- Russian is quite clear (Haiman 1983:796)

Russian

On	utomil	sebja	On	utomil+sja	
he	exhausted	REF	he	exhausted+MID	

'He exhausted himself.' — 'He grew weary.' (spontaneous event)

In Buan, supine and infinitive forms of the verb have different markers, GIVEN the same verb (p.444; table 78) and MID:

lok-6-a [lɔʔχɓa]
go.up-MID-SUP
'in order to go up'

lok-6-um-[lɔʔχɓum]
go.up-MID-INF
'to go up'

– This means categorically that MID targets root forms, maybe stem forms too, but not finite forms.

– IT IS INDEED VOICE

Table 79 seems to suggest that

- Some lexical verbs are lexically u/ɮ: their semantics allow reciprocity, body actions, emotion, cognition
- others are derived from
  - $S_{nf}\backslash NP$ (intransitive conjugation)

nonfinite

  - $(S_{nf}\backslash NP)\backslash NP$ (Transitive conjugation)

Look at REL!

- From the previous slide, Bunan does harness the verb form distinction.

- Voice has this robust property that it wants to see all arguments of the verb before doing something.
  (e.g. the Turkish causative: DAT or ACC?)
- Categorically, this means it targets IV, not VP or S/NP, since the last two can be syntactically derived.
- Turkish, Bunun, Paiwan: Their voice systems must together say something. (for UDK 2027!!)
- For Bunun: IV, VP are S_{nf}\NP

Genuine middle verbs (p.458)

- All of these have the transparent morphology of V-6-um

  to walk

  to become insane

  to start going

  ⋮

  to inhale

- I don't see any reason why they should be called lexically middle verbs.

1) Forms of the verb (p. 478–490)

		INTR	MID	TR
– Finite form		√-ek	√-ek	√-kata (FUT)
– The infinitive		√-men	√-ɸom	√-tɸ-om
– The supine (aka. INF of purpose)		√-de	√-ɕ-a	√-tɕ-a
– Participle	progressive	√-ka	√-ɕ-ka	√-ka (SG)
	active	√-i	√-ɕ-i	√-tɕ-i
– The converb		√-tɕʰi	√-ɕ-tɕi (↑ only derivation allowed)	√-ɸ-dʑi (PL)

finite MID

– ek :: $(S_f\backslash np\backslash np) \backslash\backslash (S_{nf}\backslash np\backslash np) : \lambda p \lambda x \lambda y.\ p\,x\,y \wedge closer'\,y\,x$

called extending category slash in Bozsahin 2025:69

(for correlational reference, in contrast with transparent reference)

semantically transitive

infinitive MID

– 6-um :: $(S_{nf}\backslash np\backslash np) \backslash\backslash (S_{nf}\backslash np\backslash np) : \lambda p \lambda x \lambda y.\ \underbrace{p\,x\,y}\ \wedge closer'\,y\,x$

↳ perhaps inf?

Transitive vs. middle conjugation (p. 461)

ɓuk_tɓ_um 'to comb someone's hair'
TR

ɓuk_ɓ_um 'to comb one's hair'
MID

ɓuk :: $S_{nf}\backslash NP_{erg}\backslash NP_{abs}$: $\lambda x \lambda y.\ comb' xy$

ɓuk_ɓ_um $(S_{nf}\backslash NP\backslash NP_{abs}$: $\lambda x \lambda y.\ comb' xy$ $\quad (VP|NP)\backslash\backslash(IV|NP)$: $\lambda p \lambda x \lambda y.\ p\,xy \wedge closer' y x$ $\Rightarrow$ $VP|NP$ $\lambda x \lambda y.\ comb\,xy \wedge closer' y x$

(An example from p. 462)

(-ek is finite form)

two arguments

(545) gi taJ kra ɕuk-tɕ-ek

1SG 3SG.GEN hair comb-TR-POS.CJ.SG

'I am combing his hair'

$s\backslash np\backslash np : \lambda x \lambda y.\mathrm{comb}'xy$

(546) gi kra

$$\dfrac{\text{ɕuk}}{S_{nf}\backslash np\backslash np \;:\; \lambda x\lambda y.\mathrm{comb}\,x\,y} \qquad \dfrac{\text{-ɕ-ek}\ \ \text{-MID}}{(s\backslash np)\backslash(s_{nf}\backslash np\backslash np) \;:\; \lambda p\lambda x\lambda y.\,p\,x\,y}$$

closer y x

$s\backslash np\backslash np : \lambda x\lambda y.\mathrm{comb}'xy$

closer x y

Syntactic type is same but semantics is different:

IV) Two types of relative clauses (sound familiar?)

(p. 755)

— Figure 44 explains it well:

- nominalized verb form
  - active -i
    - agentive nominalizer -pa — FINITE REL
    - others (REL) =tsuk — FINITE REL
  - infinitive -men/-om
    - (no -pa or -i): =tsuk — INFINITIVE REL

Categories of -i, -pa, =tsuk and -men are doing the syntax-semantics work of the Bunun relative clause:

- -i and -men are NOT relativizers (also -un)
- -pa and =tsuk are relativizers
- -pa seems to be insistent on finiteness on the RC.
- =tsuk leaves it underspecified.
- REL in Bunun is always extracting (from the RC on the left)

  schematically $(NP/NP)\backslash RC$

$-i$ :: $(S_f \backslash NP) \backslash (S_{nf} \backslash NP)$: $\lambda P.\ P$

nonfinite

$-pa$ :: $(NP/NP) \backslash (S_f \backslash NP_{erg?})$: $\lambda P \lambda Q \lambda x.\ Px \wedge Qx$

finite

$= tsuk$ :: $(NP/NP) \backslash (S \backslash NP)$: $\lambda P \lambda Q \lambda x.\ Px \wedge Qx$

underspecified

$-men$ :: $(S_{nf} \backslash NP) \backslash (S_{nf} \backslash NP)$: $\lambda P.\ P$

- Not sure about NP$_{erg}$: $\left\{ \begin{matrix} \text{erg} \\ \text{abs} \end{matrix} \right\}$ are pragmatically controlled (p. 213)
  - erg required in transitive conjunction

Schematically we have the following syntactic structure of the RC and the modified NP :

[RC]	-i	=tsuk	$\longrightarrow$ -i=tsuk	NP/NP
[RC]	-i	-pa	$\longrightarrow$ -i-pa	NP/NP
[RC]	-men	=tsuk	$\longrightarrow$ -men=tsuk	NP/NP
*	-men	-pa		*

$\underbrace{\text{RC}}_{S_{nf}\backslash NP}$ $-i$ $-pa$ $\longrightarrow NP/NP$

$S_{nf}\backslash NP \quad (S_f\backslash NP)\backslash(S_{nf}\backslash NP) \quad (NP/NP)\backslash(S_f\backslash NP_{erg?})$

$\underbrace{S_{nf}\backslash NP \quad (S_f\backslash NP)\backslash(S_{nf}\backslash NP)}_{S_f\backslash NP}$

FINITE because of -i, -pa

$\underbrace{\text{RC}}_{S_{nf}\backslash NP}$ $-i$ $=tsuk$ $\longrightarrow NP/NP$

$S_{nf}\backslash NP \quad (S_f\backslash NP)\backslash(S_{nf}\backslash NP) \quad (NP/NP)\backslash(S\backslash NP)$

FINITE because of -i

$\underbrace{S_{nf}\backslash NP \quad (S_f\backslash NP)\backslash(S_{nf}\backslash NP)}_{S_f\backslash NP}$

underspecified: erg or abs

$\underbrace{\text{RC}}_{S\backslash NP}$ $-men$ $=tsuk$ $\longrightarrow NP/NP$

$S\backslash NP \quad (S_{nf}\backslash NP)\backslash((S_{nf}\backslash NP) \quad (NP/NP)\backslash(S\backslash NP)$

INFINITIVE

underspecified: tensed or untensed

Criticallly,

* -men -pa. This is how we eliminate it:

$$\underbrace{\text{RC}}_{S\backslash NP}\quad \text{-men}\quad \text{-pa}$$

$$\underbrace{S\backslash NP \quad (S_{nf}\backslash NP)\backslash(S_{nf}\backslash NP)}_{S_{nf}\backslash NP} \quad (NP/NP)\backslash(S_f\backslash NP_{arg?})$$

$$\underbrace{S_{nf}\backslash NP \quad (NP/NP)\backslash(S_f\backslash NP_{arg?})}_{***}$$

(also, for -um)

- There is no role asymmetry in Bunan RCs.

$\Longrightarrow$ IV, VP are not doing any work in Bunan constructions.

- $S\backslash NP$ is either finite, $S_f\backslash NP$, or non-finite, $S_{nf}\backslash NP$.

- why? Because the '\NP' can be anything unless further restricted to be agent (verbs?)
(for -po on transitive conjugation)

- Can -i- pa extract $\mathcal{S}_{(0)}$? (1166-1171) apparently NO. ((1172): NP must be agent (not exp or abs))

- Can -i=tsuk extract $\mathcal{S}_{(0)}$? YES (1176)

- Can -men=tsuk extract $\mathcal{S}_{(0)}$? YES (1198)

$\Longrightarrow$ no role asymmetry in relativization. Just agent restriction in one case.